

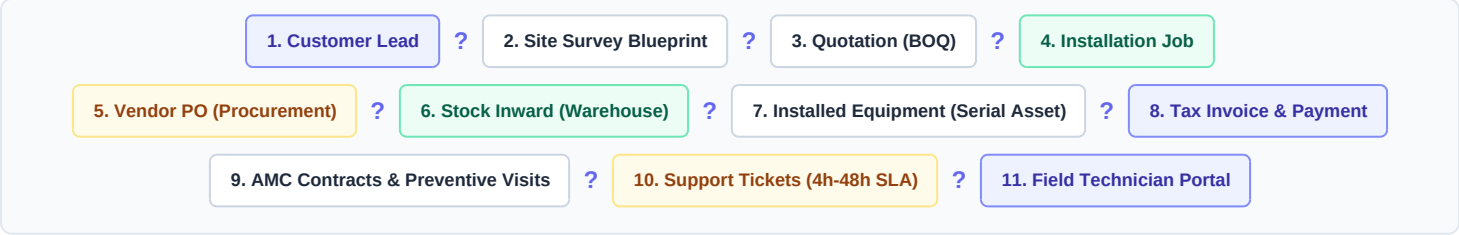
CCTV CRM SYSTEM ARCHITECTURE

OFFICIAL SYSTEM SPEC

Enterprise Module Data Flow & Real-Time Processing Specification

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1. MASTER SYSTEM DATA FLOW OVERVIEW



2. SALES PIPELINE, SURVEYS & QUOTATION ENGINE

MODULE	PRIMARY TABLES	INPUT & TRIGGERS	DATA PROCESSING & OUTPUT STATE
Leads Management	leads lead_histories	Phone / WhatsApp / Web Inquiry / Walk-in	Stages: new ? contacted ? site_visit_scheduled ? quoted ? won/lost. Immutable audit trail in lead_histories.
Site Survey Assessment	site_surveys site_survey_photos	Technician Site Inspection	Stores indoor/outdoor camera counts, conduit & cable meters, NVR storage retention days, and physical site blueprint photos.
Quotation (BOQ) Engine	quotations quotation_items	Survey specs + Product catalog	Auto-calculates item totals (qty × price), subtotal, discount deductions, 18% GST tax, and grand total. Generates QUO-YYYYMMDD-XXXX.

3. FIELD OPERATIONS, INSTALLATION & HELPDESK SLA

MODULE	PRIMARY TABLES	INPUT & TRIGGERS	DATA PROCESSING & OUTPUT STATE
Installation Jobs	installation_jobs	Approved Quotation (Quote Won)	Dispatches technician; tracks hardware mounting, wiring, IP configuration. Status: pending ? assigned ? in_progress ? completed.
Installed Equipment Registry	installed_equipment	Job Completion Handover	Registers hardware serial numbers, static IP addresses, camera locations (e.g. 'Main Gate'), installation date, warranty calculation.
AMC Contracts & Visits	amc_contracts amc_visits	Customer Service Agreement	Generates contract (AMC-XXXX) and auto-schedules preventive maintenance visits (e.g. 4 quarterly visits). Tech records lens cleaning & UPS test.
Support Tickets (SLA)	service_tickets	Client Fault / Offline Camera	Priorities: Critical (4h), High (12h), Medium (24h), Low (48h). Auto-links AMC/Warranty coverage. Tech logs troubleshooting & replaced parts.

4. WAREHOUSE INVENTORY, PROCUREMENT & INWARDING LEDGER

Automatic Inwarding Guarantee: Marking a Purchase Order (PO) item received increments physical stock in `products.stock_quantity` and writes an immutable audit record in `stock_movements` with type `'in'`.

MODULE	PRIMARY TABLES	INPUT & TRIGGERS	DATA PROCESSING & OUTPUT STATE
Suppliers & Vendors	suppliers	Distributor Registration	Stores vendor contacts, GSTIN, payment terms (Net 30, Advance), and aggregates cumulative purchase volume.
Purchase Orders (PO)	purchase_orders purchase_order_items	Stock Replenishment Order	Generates PO-YYYYMMDD-XXXX. Calculates unit cost, subtotal, 18% GST tax, shipping. Status: draft ? ordered ? received.
Warehouse Stock & Movements	products stock_movements	PO Inward / Job Consumption	Real-time inventory levels, minimum stock alerts. Logs every stock movement (type, quantity, balance_after, user_id).

5. INVOICING, GST BILLING & PAYMENT COLLECTION

MODULE	PRIMARY TABLES	INPUT & TRIGGERS	DATA PROCESSING & OUTPUT STATE
GST Tax Invoices	invoices	Completed Installation Job	Generates INV-YYYYMMDD-XXXX with itemized hardware line items, taxable amount, 18% GST, due date, and payment instructions.
Payments Ledger	payments	Advance / Settlement Receipts	Records transaction (Cash, Bank Transfer, UPI, Cheque, Reference No). Automatically recalculates <code>amount_paid</code> , <code>balanceDue()</code> , and status (<code>unpaid</code> , <code>partially_paid</code> , <code>paid</code> , <code>overdue</code>).

6. ROLE-BASED ACCESS CONTROL (RBAC) & SECURITY MATRIX

SYSTEM MODULE / FEATURE	ADMINISTRATOR	STAFF / SALES	TECHNICIAN	CUSTOMER
Leads & Quotations Management	FULL CRUD	VIEW & QUOTE	NO ACCESS	NO ACCESS
Field Work (Jobs, Tickets, AMC Visits)	FULL CRUD	ASSIGN & VIEW	ASSIGNED ONLY	NO ACCESS
Invoicing & Financial Ledgers	FULL CRUD	CREATE & PAY	NO ACCESS	NO ACCESS
Inventory & Vendor Procurement (POs)	FULL CRUD	CREATE & INWARD	NO ACCESS	NO ACCESS
User Accounts & Role Permissions	FULL CRUD	NO ACCESS	NO ACCESS	NO ACCESS